

Convolutional Codes & Viterbi Algorithm

Prepared by
Prof. Hitesh Dholakiya

Modified by
Rifat Rahman
Assistant Professor
Institute of Appropriate Technology, BUET

Convolutional Code

- ❑ A convolutional code is a type of channel coding (error control coding) technique used in digital communication systems to detect and correct errors introduced by noise in the channel.
- ❑ Unlike block codes (e.g., Hamming code), convolutional codes process data continuously, not in fixed-size blocks.
- ❑ Input data bits are passed through a shift register (memory element)
- ❑ Output bits are generated using modulo-2 addition (XOR)
- ❑ Output depends on: Current input bit and Previous input bits (memory). So, the encoder has memory.
- ❑ Because the output sequence is the convolution of the input sequence with the generator sequence (in modulo-2 arithmetic).

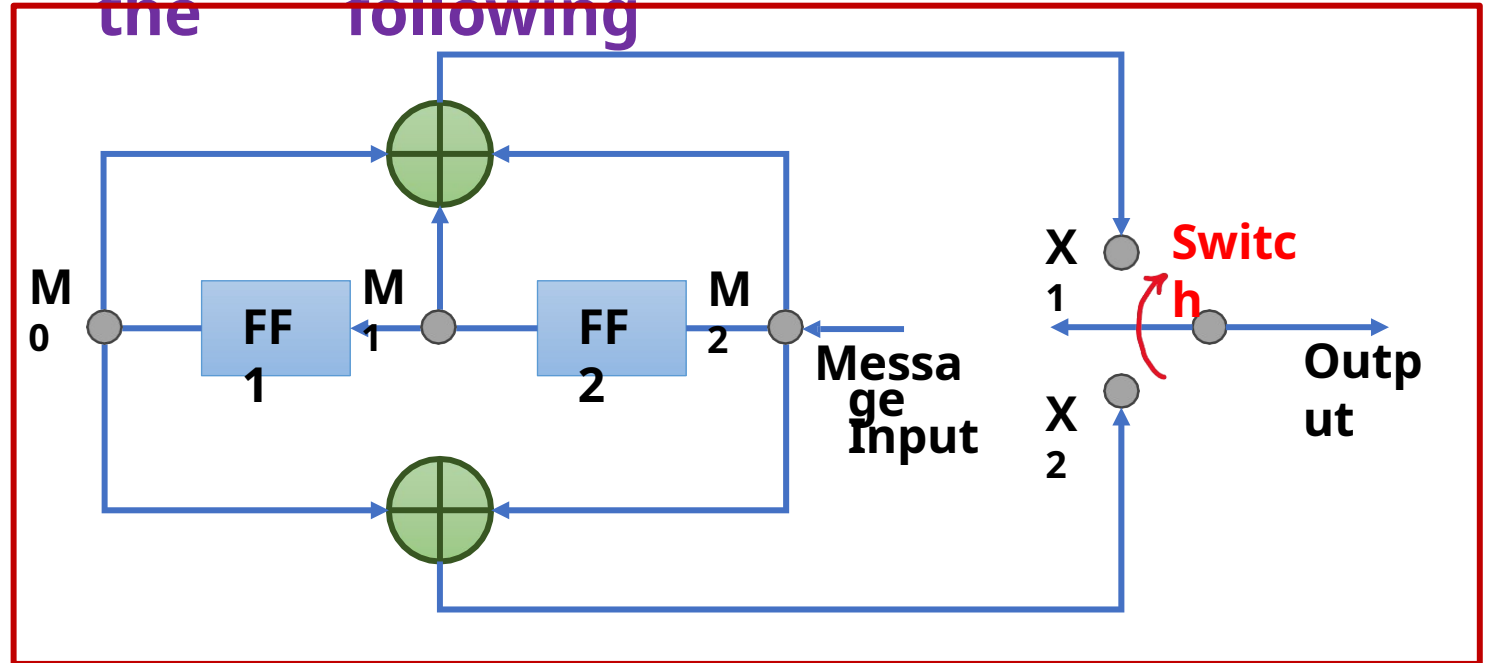
Convolutional

Example: The Convolutional Encoder is given in the figure. Calculate the following

- Constraint Length K
- Code Dimensions (n, k)
- Code Representation (n, k, K)
- Code Rate r
- States
- Output with Message $[1 \ 1 \ 0]$
- Code Tree
- Code Trellis
- State Diagram

$$\Rightarrow K = m + 1 = 2 + 1 = 3$$

↑
memory elements



In this figure,

The generator sequence for output X_1 is, $g_1 = (1, 1, 1)$

The generator sequence for output X_2 is, $g_2 = (1, 0, 1)$

→ Code Dimensions $(n, k) = (2, 1)$

– n = Number of Outputs at time

k = Number of Inputs at time

→ Code Representation

$$\Rightarrow (n, k, K) = (2, 1, 3)$$

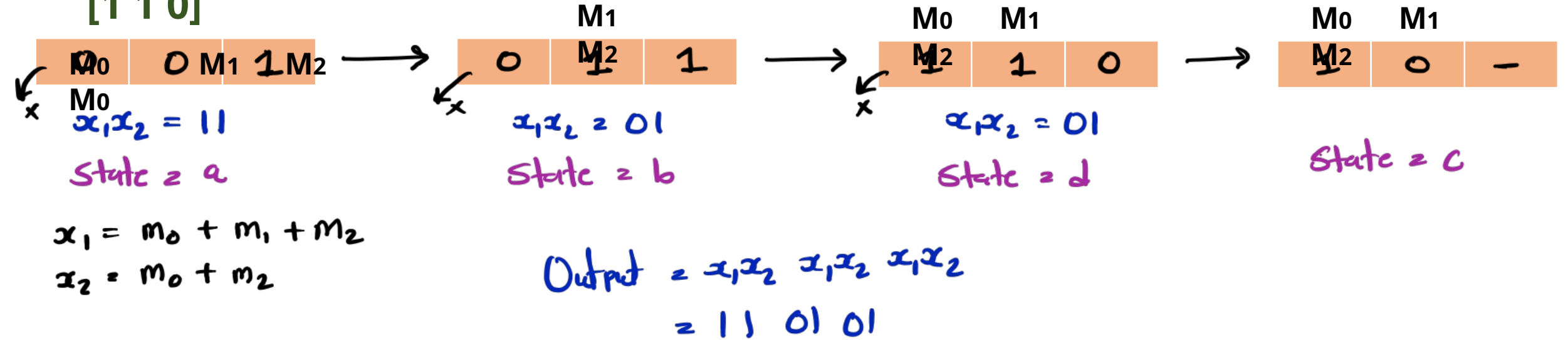
→ Code Rate γ

$$\Rightarrow \gamma = \frac{k}{n} = \frac{1}{2}$$

■ **Stat** (based on memory elements)

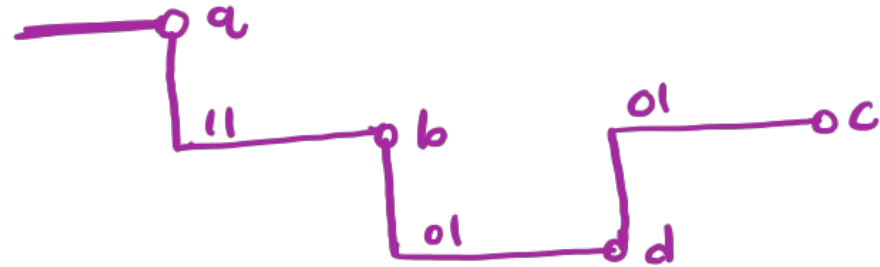
m_0	m_1	State
0	0	a
0	1	b
1	0	c
1	1	d

Output with Message [1 1 0]



Code Tree

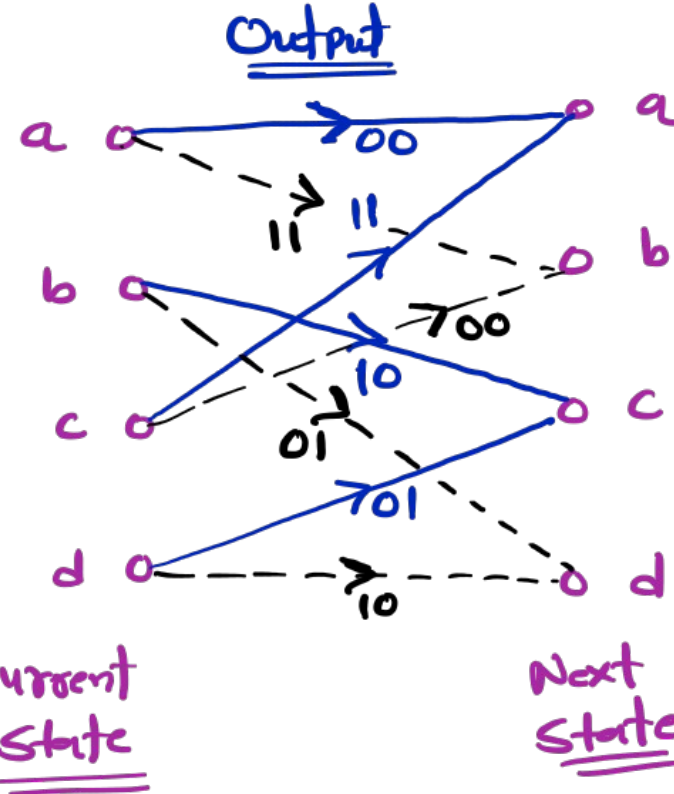
~ If Input = 1 → Step down
 Input = 0 → Step Up



Code Trellis and State Diagram

		m_0m_1		m_2m_2		
M_0	M_1	M_2	X_1	X_2	Current State	Next State
0	0	0	0	0	a	a
0	0	1	1	1	a	b
0	1	0	1	0	b	c
0	1	1	0	1	b	d
1	0	0	1	1	c	a
1	0	1	0	0	c	b
1	1	0	0	1	d	c
1	1	1	1	0	d	a

Code Trellis



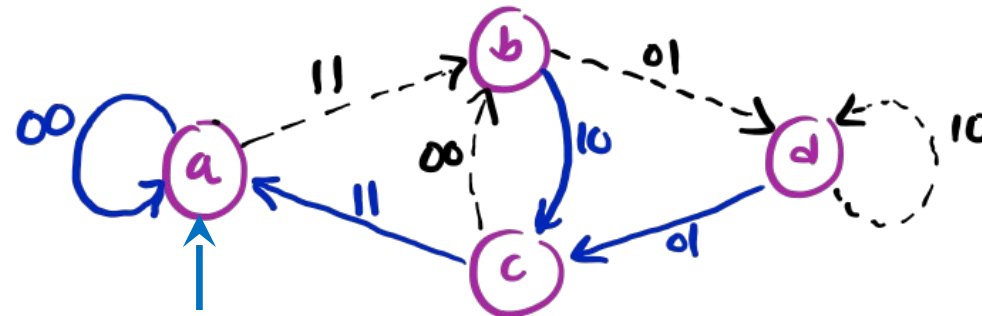
Input 0

Input 1

State Diagram

Input 0

Input 1



Viterbi Algorithm

Outline

- es ☐ Basics of Viterbi Algorithm
- ☐ Objective of Viterbi Algorithm
- ☐ Process of Viterbi Algorithm
- ☐ Example of Viterbi Algorithm

Basics of Viterbi Algorithm

- ❑ It is a maximum-likelihood decoding method for convolutional codes.
- ❑ A trellis diagram is used to decode the received sequence.

Objective of Viterbi Algorithm

- ❑ To find the most likely transmitted bit sequence by selecting the path through the trellis with the minimum path metric (distance from the received sequence).

Process of Viterbi Algorithm

- ❑ The following steps are used to apply the Viterbi Algorithm:
 - **Construct the trellis diagram** from the convolutional encoder.
 - **Initialize path metrics** (usually zero for the starting state and infinity for others).
 - **Compute branch metrics** by comparing the received sequence with expected outputs.
 - **Update path metrics for each state** (Add-Compare-Select operation).
 - **Store survivor paths for each state.**
 - **Trace back through the trellis** to obtain the decoded sequence.

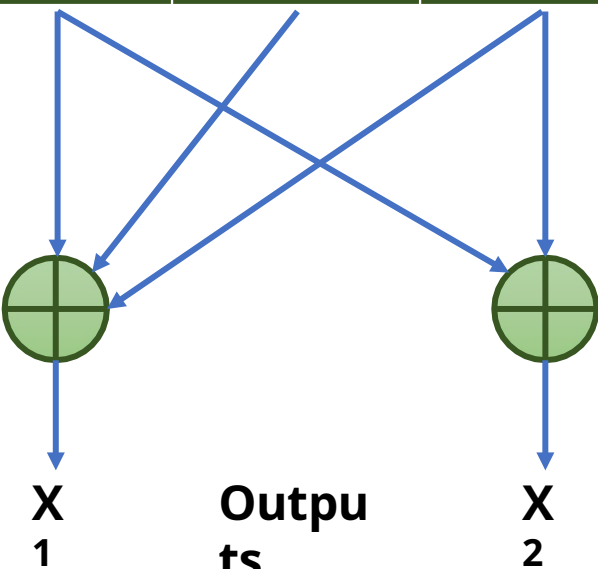
Example of Viterbi Algorithm

Example: Identify the decoded data using the Viterbi Algorithm if the received sequence is 11 00 00.



Input

Input = 0
Input = 1



Outputs

$$x_1 = m_1 + m_2 + m_0$$

$$x_2 = m_0 + m_2$$

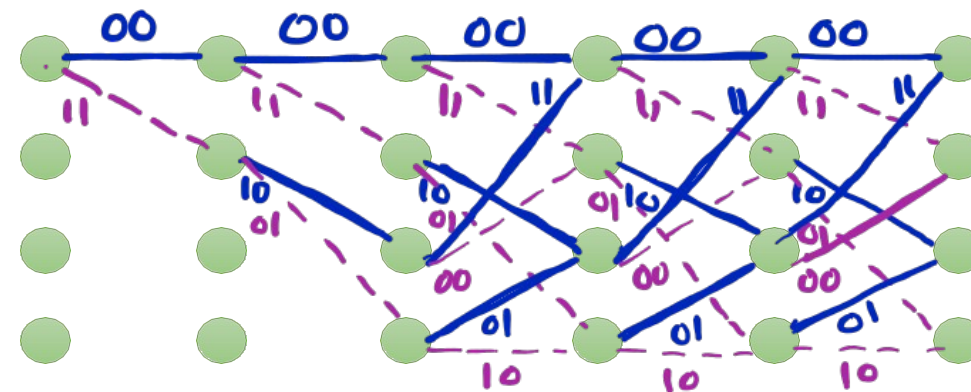
M0	M1	M2	X1	X2	Current State	Next State
0	0	0	0	0	a	a
0	0	1	1	1	a	b
0	1	0	1	0	b	c
0	1	1	0	1	b	d
1	0	0	1	1	c	a
1	0	1	0	0	c	b
1	1	0	0	1	d	c
1	1	1	1	0	d	d

State a [00]

State b [01]

State c [10]

State d [11]

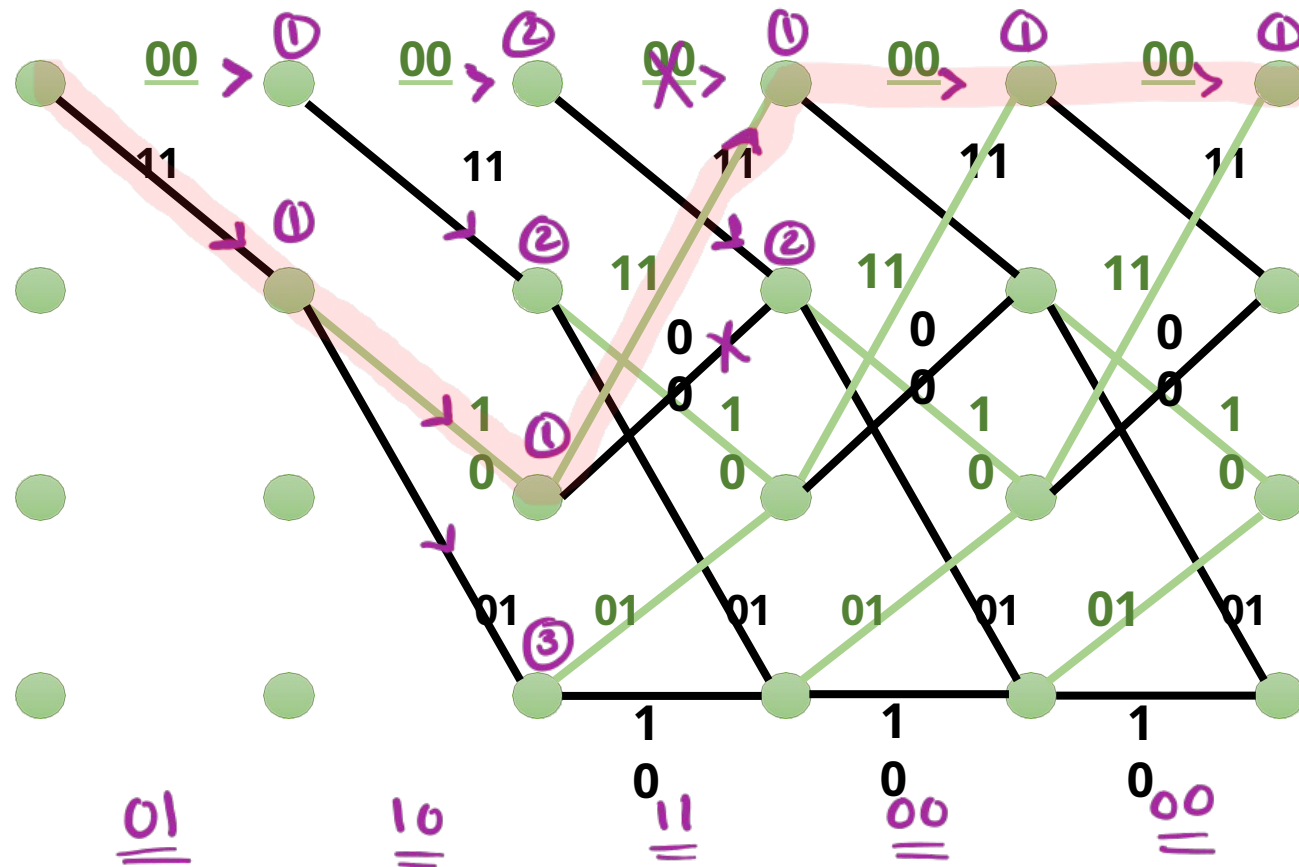


State a [00]

State b [01]

State c [10]

State d [11]



⇒ Decoded data = 01 10 11 00 00
 ↑
 One bit error